

PHASE 1 — DIAGNOSTIC CHARACTERIZATION

Phase 1a: Collapse Characterization

EM 0.62→0.34, monotonic decline

Target: slow the slope

Phase 1b: SVD Geometry



RL 2–6× aligned; SFT 1.47×

Descriptive weight-geometry correlate

Phase 1c: Activation Probing



No activation signal (negative)

Motivates causal intervention in Phase 2

PHASE 2 — DISTILLATION & CAUSAL INTERVENTION

DPO vs SFT

+1.0pp (n.s.)

Preference alignment insufficient

Truncation Confound

Earlier +2.4pp was artifact

Controlled re-evaluation needed

Oracle Retrieval

+47.3pp recovery ($p < 0.0001$)

Retrieval API is the key lever

Conclusion

Behavioral Mismatch ($\sim 87\%$) $\gg$ Capacity ($\sim 13\%$)

Student learns search strategy but cannot execute without retrieval API